Zheyan Shu

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Education

Peking University, BSc. in Physics — Sept. 2022 – July 2026 (Expected)

- GPA: 3.63/4.0 (Cumulative), 3.66/4.0 (Major)
- **Advanced Courses:** Quantum Field Theory, Quantum Gauge Field Theory, General Relativity, Quantum Statistical Mechanics, Group Theory II, Solid State Theory
- **Self Studied:** Modern Quantum Mechanics (J.J. Sakurai), String Theory I (J. Polchinski), Supersymmetry and Supergravity (J. Wess and J. Bagger)

Research Interests

Quantum field theory and its applications, string theory, holography and quantum gravity

Research Experience

Research on Black Holes and Fortuity — Nov. 2025 – Present
Advisor: Prof. Chi-Ming Chang — YMSC, THU

- *Project: Fortuitous states in the 3d $\mathcal{N} = 2$ critical $O(N)$ model*
  - Computed the superconformal index and analysed the spectrum to identify and characterise fortuitous states.
- *Project: Fortuitous operators in monopole sectors of ABJM theory*
  - Constructing explicit fortuitous operators in monopole sectors via the state-operator correspondence.

Undergraduate Research on Conformal Field Theory and Collider Physics — May 2024 – May 2026
Advisor: Prof. Hua Xing Zhu — School of Physics, PKU

- Conducted literature review on advanced topics including the conformal collider and ANEC.
- *Project: Calculation of energy-energy correlation (EEC) at Leading Order (LO)* — *June – Aug. 2024*
  - Manually calculated the LO angular distribution using Feynman diagrams.
  - Devised a proper treatment of the plus distribution, obtaining the correct LO EEC from angular correlations.
- *Project: Angular correlations in dihadron production at NLO* — *Oct. 2024 – June 2025*
  - Computed phase-space integrals of the hard process using IBP reduction and differential equations.
  - Performed resummation in the unphysical region and analysed singular behaviour in the collinear and soft limits.
- *Project: Light-ray operators and collider correlation functions* — *Oct. 2025 – May 2026*
  - Explored analogues of light-ray operators in QCD and their physical applications in collider physics.
  - Investigated possible connections between charge-charge correlations (QQC) and open quantum systems.

Publications

Dihadron Angular Correlations in the e^+e^- Collision — Sept. 2025
W.-L. Ju, ***Z.-Y. Shu***, T.-Z. Yang, Z.-H. Zhang, and H. X. Zhu
J. High Energ. Phys. 2025, 51 (2025). arXiv:2506.11463

Awards

Outstanding Undergraduate Research Project (2024)

Academic Activities

Study on Fundamental Aspects of Quantum Mechanics and QFT *(Organizer & Presenter)* — 2024
Advisor: Prof. Yinan Wang — School of Physics, PKU

- Led a group of fifteen undergraduate students in a review of foundational problems in quantum mechanics.
- Delivered two presentations, one on path integrals and one on quantum chaos .

Third Workshop on Frontiers in QFT — Non-Perturbative Methods *(Participant)* June 2025

First Joint Summer School on Theoretical High-Energy Physics *(Participant)* Aug. 2025

5th Symposium on Quantum Field Theory and its Applications *(Participant)* Oct. 2025

New Frontiers of Quantum Field and Gravity *(Participant)* Jan. 2026

Teaching Experience

Teaching Assistant, *Introduction to Modern Physics I*, PKU, Fall 2025

Teaching Assistant, *Group Theory II* (graduate level), PKU, Spring 2026

Skills

Computing: C++, Python, LaTeX, Mathematica

English: TOEFL iBT 108/120 (R30, L27, S24, W27)